

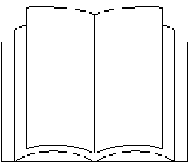


Operator's manual for Attachment



Forklift Mast

Product number: 400-700-series loaders: A32915



Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

Manufacturer:

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This manual uses the following warning symbols to highlight factors that must be taken into account to reduce the risk of personal injury or property damage:

	DANGER:
	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or property damage. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.



Note:

This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. In some areas the use of the pallet forks, or any other kind lifting equipment, may require personnel who have received special training and who are authorized to use the equipment. Find out the applicable regulations in your area before putting the forklift mast into service.

Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This operator's manual is a translation of the original Finnish version.

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this AVANT attachment. It has been designed and manufactured on the basis of years of product development and experience. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

With these instructions, even an inexperienced user can use the loader and the attachment safely. The manual includes important instructions for experienced AVANT operators as well. Ensure that all persons using the equipment have received proper guidance and familiarised themselves with the manual and all safety instructions before using the equipment. Keep this manual at hand throughout the service life of the equipment. If the manual is lost, you can request a new one from the retailer or manufacturer. If you sell or transfer the equipment, be sure to hand over this manual to the new owner.

2. Designed purpose of use

The AVANT forklift mast is an attachment suitable for use with AVANT multi purpose loaders. The forklift mast gives more lift capacity and stability with large loads. The forklift mast is an ideal attachment for handling and lifting of heavy pallets (stone, fertilizer) and other heavy materials. It can be also used for loading and unloading of trucks. With the fork side shift mechanism, pallets of different sizes can easily be transported.

The mast is equipped with rigid, foam filled tyres, which gives good all terrain transporting capabilities. Thanks to the quick attach system connecting to and from the Avant is quick and easy. Forklift mast can be mounted only on machines that are equipped with telescopic boom. The forklift mast must not be used for lifting persons and it is prohibited to use the attachment to lift any kind of access platform.

The attachments have been designed to require as little maintenance as possible. Regular maintenance tasks can be performed by the operator. Read the maintenance and servicing instructions in this manual. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for the pallet forklift mast

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not familiarised yourself with its operation and the related hazards.

The purpose of these instructions is to help you to:

- use the equipment in a safe and efficient manner
- monitor and prevent any hazardous situations
- keep the equipment intact and ensure a long service life



- When mounting the pallet forks to the loader, ensure that the locking pins of the loader's quick attach plate are **in the lower position** and that they have **locked** the attachment **to the loader**.
- Move the loader controls calmly and deliberately to prevent the load from falling.
- Be especially careful when handling elevated material.
- Always transport the load as low and close to the machine as possible to keep the centre of gravity low.



- Always keep the tyres of the forklift mast on the ground or close to the ground surface. When handling loads with the forklift mast, its tyres must be on the ground.
- Never leave the driver's seat when the forks have been elevated. Lower the forks on ground and keep the tyres of the mast on the ground.
- Shut down the loader before any maintenance or adjustments.
- Pay attention to the contours of the terrain and other hazards, such as branches and trees reaching the driver's area. Pay attention to the surroundings and any other persons and machines moving in the vicinity.
- On inclined terrains drive carefully and keep the load as low as possible.
- Never go under the raised load or the loader boom.
- The forklift mast, or the loader, is not intended to be left to carry loads for longer periods. It must be observed that the loader boom and the forks will lower by themselves.
- Using the forklift mast to lift persons or as an access platform is strictly prohibited.
- Do not modify the structure of the attachment in a manner that would affect its safety. Welding, drilling or any other modification of the pallet forks is prohibited.
- Read also the safety instructions for the loader from the operator's manual of the loader.



Familiarise yourself with the controls of the loader in a safe area. Pay particular attention to the increased turning radius when the forklift mast is attached.



Note that the forks of the forklift mast may be lowered or tilted even if the engine has been stopped. Never go under or in front of a raised attachment or the loader boom. Do not leave the mast elevated.

Remember to wear proper protective clothing:



- The noise level in the driver's seat exceeds 85 dB(A). Wear hearing protection while working with the loader.



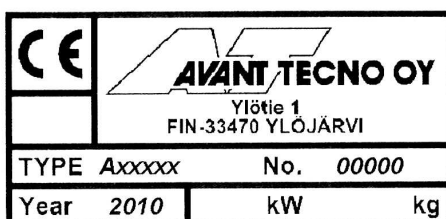
- It is recommended to wear safety boots working with the loader. Also wear protective gloves.

4. Forklift mast specifications

Listed below are the markings and labels fastened to the pallet forks, which must be visible on the equipment. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer or contact information provided on the cover page.



The warning decals contain important safety information. Replace damaged or missing decals with new ones.



- A46771 – Read instructions before use.
- A46772 – Do not go under a raised attachment, stay away from raised equipment.

Identification plate of the attachment

Table 1 – Forklift mast Specifications

Product number	A32915
Total weight:	300 kg
Lift height:	1 750 mm
Fork length:	1 100 mm
Fork separation:	330 - 470 mm
Fork width:	100 mm
Maximum working load:	1500 kg
Suitable AVANT loader*:	400-700

The forklift mast is equipped with a hose burst valve.

*The forklift mast can only be used with loaders equipped with a telescopic boom



A46771



A46772

5. Assembly of the forklift mast

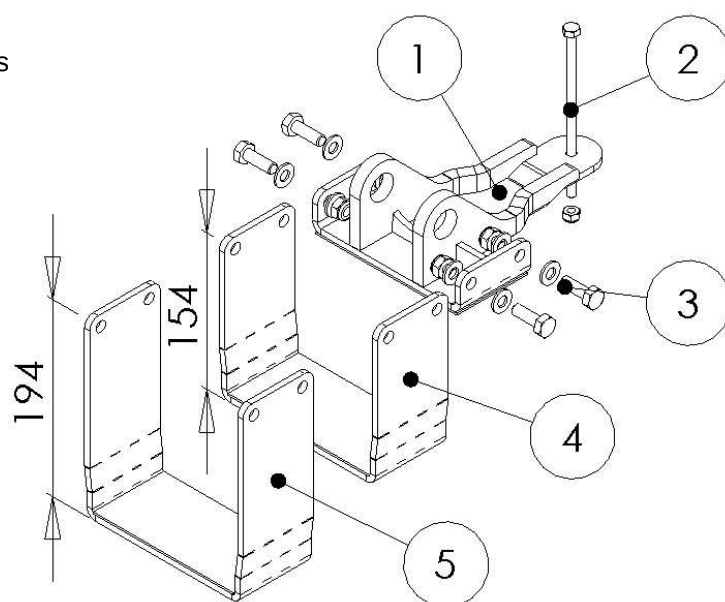
As the forklift mast is delivered, for easier transportation the forks and the assembly kit are separated. The forklift mast can be assembled and put into service easily without assistance. After the forklift mast and its mounting kit are assembled for the first time, it is quick and easy to mount the mast to the loader. When assembling the forklift mast remember correct working methods, as the components are heavy. A few basic tools are needed for assembly.

5.1. Mounting the mounting kit to the loader

To mount the forklift mast, a mounting bracket must be mounted on the loader boom to fit the supporting rod of the forklift mast. A mounting kit A35039 is delivered with the mast, with which it is possible to mount the forklift mast on 400-700 series loaders.

The mounting kit contains the following components

1. Mounting bracket
2. Screw and nut for the mounting bracket
3. Screws (4), washers (8) and nuts (4) for the lower bracket.
4. Lower bracket for 400-600-series loaders
5. Lower bracket for 700-series loaders (2011-)

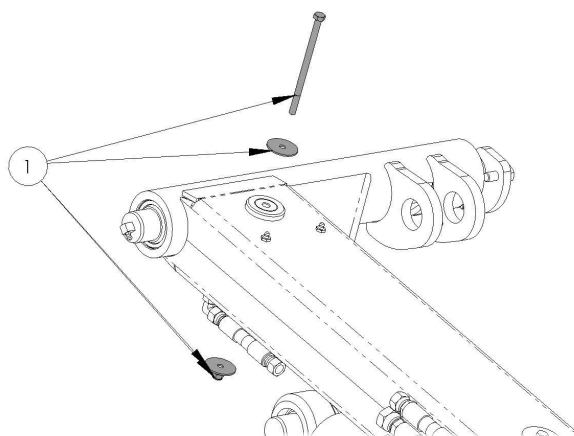


To make the mounting of the kit easier, lift the loader boom and apply the boom service support. Shut down the loader and release the residual pressure. Take up the items shown in the figure above.



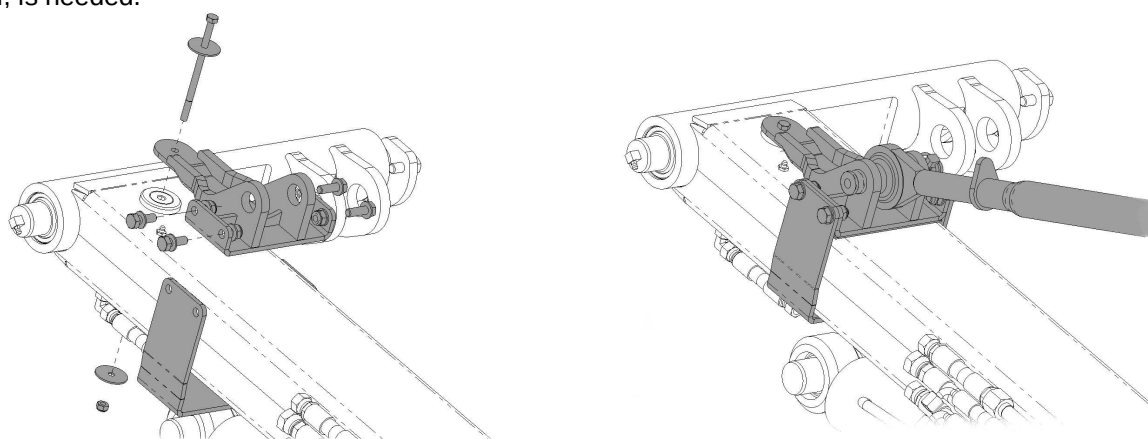
Use the service support of the boom while mounting the mounting kit on the boom.

Remove the pin holder screw from the upper part of the boom, shown in the adjacent figures.



Place the mounting bracket of the forklift mast on the boom and, if necessary, use the with the longer screw provided with the mounting kit to fasten the bracket.

Mount the correct lower bracket (part 4 or 5) and tighten the screws securely. Two lower brackets are included in the mounting kit for the forklift mast, but only one of them, depending on the type of the loader boom, is needed.



5.2. Assembling and mounting the forklift mast

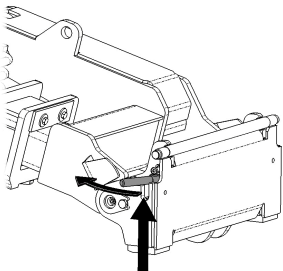
The forklift mast should be mounted on the loader before mounting the forks. The forklift mast is delivered on the pallet so the quick attach plate of the forklift mast can be reached and locked directly with the loader boom. The forklift mast should not be lifted by hand or by other means due to its weight and the risk of falling over.

These instructions are applicable to mounting the attachment to the boom quick attach plate of AVANT loaders that have standard quick attach locking. If your machine is equipped with a hydraulic locking system, refer to the relevant manuals for additional mounting and safety instructions.

Mounting the forklift mast to the loader is quick and easy, but it must be done carefully. An attachment that has not been locked may come off the loader and cause hazardous situations. Also, when mounting the forklift mast, it should be kept in mind that there is no need to lift it at any phase during assembly, usage or storing. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions in Section 3 and the instructions provided in the loader manual.

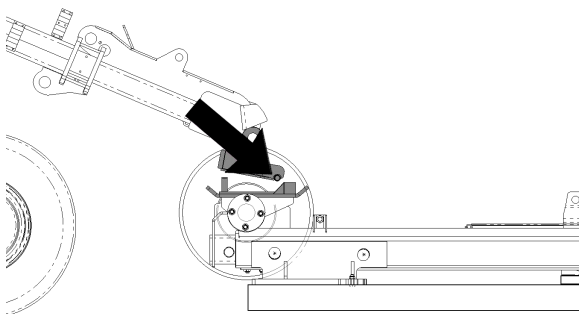


Never move or raise an attachment that has not been locked.



Step 1:

- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- Ensure that the hydraulic hoses are not in the way during mounting.

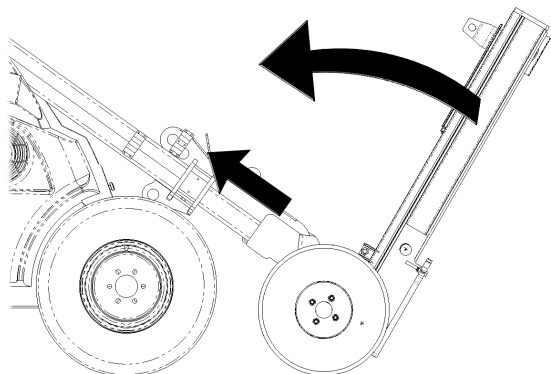
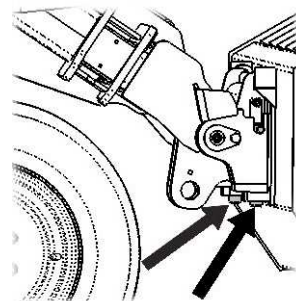


Step 2:

- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive and tilt the loader boom to the quick attach plate of the attachment. Use the telescopic boom of the loader to reach the attachment plate.
- Turn the quick attach plate completely against the attachment plate.
- **Lock the locking pins manually before lifting or tilting the attachment.**



Do not use the automatic locking of the locking pins. The forklift mast can easily tip over if tilted when unlocked. Ensure that both locking pins are at the lower position and come through the fasteners before moving or lifting the attachment.

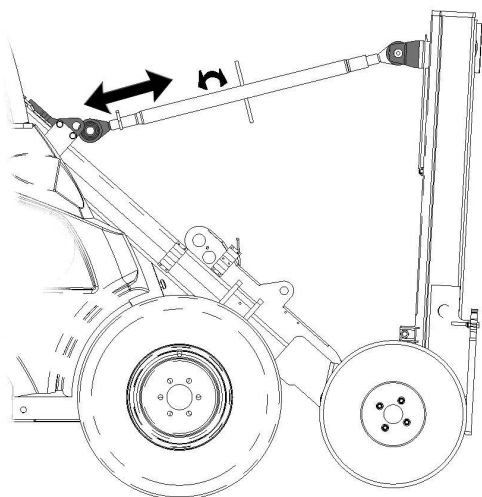


Step 3:

- Lift and tilt the forklift mast to a forward facing tilted position. Do not tilt the forklift mast to a vertical position because due to the hinged quick attach plate the forklift mast could unexpectedly tip over towards the loader.
- Keep the wheels of the forklift mast against the ground or near the ground surface.



The quick attach plate of the forklift mast is not mounted rigidly to the frame of the forklift mast. Be careful not to tilt the forklift mast too far up due to the hazard of tipping over. Mount the supporting bracket (step 4) with which the tipping is prevented and with it the forklift mast can be safely pulled to the operating position.



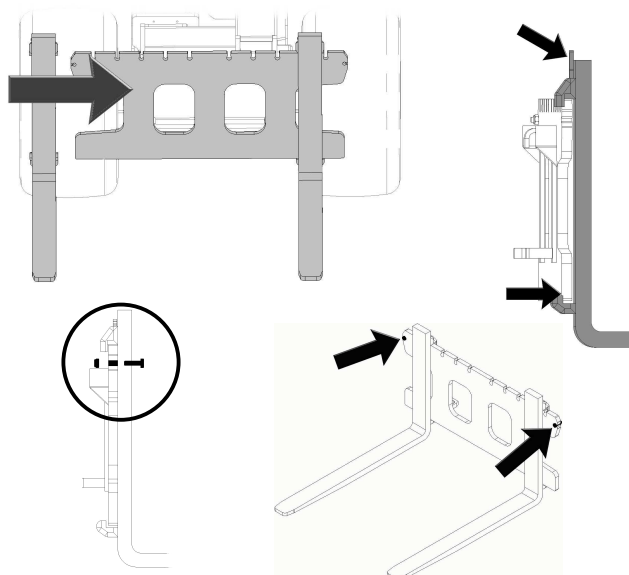
Step 4: Mounting the supporting bracket

- Shut down the loader engine. Adjust the supporting bracket to its longest position. Mount the bracket to the forklift mast and secure it with the cotter pin.
- Mount the other end of the bracket to the loader. Do not use your fingers to check the alignment and watch out not to tip over the forklift mast. Stay clear from the area between the mast and the loader boom.
- After mounting the bracket, it should be adjusted so that the forklift mast is vertical when its wheels are close to the loader.

5.2.1. Mounting the forks

After the forklift mast is mounted on the loader, the forks can be slid to the frame. Turn the locking pin on top of the fork to the upright position, slide the forks on the frame and lock them to one of the locking positions. Make sure that also the lower bracket is fixed onto the frame. Note that the components are heavy; remember healthy working habits.

Along with the mounting kit two M12 bolts and nyloc nuts are provided to prevent the forks from accidentally sliding off from the frame. The bolts are fastened to the holes on the sides of the frame as shown in the figures. Tighten the bolts securely.



5.3. Connecting and disconnecting the hydraulic fittings

Before connecting or disconnecting hydraulic fittings, shut down the loader engine and move the auxiliary hydraulics lever on the loader to its extreme positions a few times. This way, there will be no remaining pressure left in the system and the fittings are easy to connect.

To connect and disconnect the fittings, move the collar at the end of the female fitting. The hoses should be connected so that the fittings with coloured caps are connected to each other. Before disconnecting the fittings, lower the forks for storage position as shown in section 5.4, shut down the loader engine and release the residual pressure, so that the fittings are easy to connect the next time the equipment is used.



When removing the attachment, always disconnect the hydraulic fittings before detaching the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.



Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure. High-pressure fluid ejection may cause injuries.

Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. Never leave the hoses hanging on the ground; place them on top of the equipment.

5.4. Taking the forklift mast off from the loader and storage position

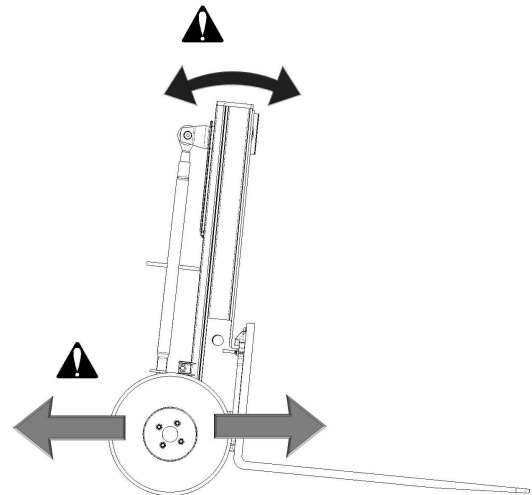
Take the forklift mast only on an even surface. Lower the wheels of the forklift mast on the ground and lower the forks to about 15-20 cm from the ground. Disconnect the hydraulic fittings and release the residual pressure. Adjust the forks to the widest setting. Disconnect the supporting bracket from the loader and set it so that the quick attachment plate can be disconnected. For temporary storage set the forklift mast to the position shown in the next page, so that the tips of the forks are against the ground and open the quick attachment locking.



When storing the forklift mast, it must be noted that the fork frame can move freely upwards. The forklift mast should not be left leaning against its forks unattended or for long periods.



Ensure the stability of the unmounted attachment. Unmounted forklift mast can slide or tip over, if the surface is not even or when working near the forklift mast. If needed, use load securing straps or similar means to secure the forklift mast from falling over, especially if it is necessary to move near the mast.



For longer storage periods the forks of the forklift mast must be removed and the forklift mast must be lowered to suitable platform, from where it can be attached as shown in section 5.2.

6. Using the forklift mast

After the forklift mast has been assembled and mounted according to instructions, it is ready for work. Read the following instructions about using, inspections and maintenance before putting the equipment into service.

- Do not use the boom floating system while the forklift mast is attached.
- Before use check that the pallet forks are intact and that all obstacles have been removed from the working area. Remember correct working methods and avoid descending from the loader unnecessarily.
- Check that the forks are positively mounted on the frame plate and that the side shift locks of the forks have been locked.
- Ensure that the loader and the attachment are used in a safe manner and as intended.
- Check the condition of the forklift mast regularly. Refer to section 7 for maintenance instructions.
- The forklift mast must not be used forks to lift persons or any kind of access platforms
- Check the condition of the pallet before lifting it. A pallet that is in poor condition or broken may cause a hazardous situation if the load may fall.



Ensure that there are no other persons at the operating area. Never go under an elevated load, or to the area between the forklift mast and the loader boom, or in directly in the front of the equipment. Also do not let others to enter the hazardous areas. Also make sure that reversing with the loader is safe.



Lower the load before leaving the driver's seat.



When lifting load from a high level, the loader could tip forward when reversing with the loader. Do not reverse the loader while simultaneously dragging the load. Make sure that the loader can handle the load being lifted before driving.

6.1. Lifting and lowering of the forklift mast

Adjust the fork width in a way that the load is distributed horizontally evenly to achieve the best possible stability. Always lock the forks to the intended gaps on the pallet fork frame.



Adjust the forks to as wide setting as possible and avoid uneven horizontal load distribution for the best horizontal stability.



Lift the load only with the lifting cylinder of the forklift mast and keep the wheels of the mast against the ground while lifting. Do not lift a loaded attachment with the loader boom.

Keep the auxiliary hydraulics of the loader at the 1-pump setting. The forklift mast is equipped with a hose burst valve, which will prevent the sudden lowering of the mast in case a hydraulic hose bursts. The valve will limit the lowering speed in normal operation as well, and it may not be possible to lower the load with full speed, as the valve reacts to high flow rates.

Lower the load by starting the lowering slowly and then gradually increasing the lowering speed. If the control lever is turned to its extreme position right away, as with the electric joystick buttons, the valve may close and stop the lowering. In this case the load should be lifted a little before continuing to lower it, so that the valve will open completely.

6.1.1. Safe working load

A maximum of 1500 kg loads can be handled with the forklift mast. Heavy loads will affect the manoeuvrability of the loader, so that with lighter loader models the following points should be noted especially on inclined terrains. The lifting capacity and the stability of the loader are at the best, when

- the loader articulation is kept straight
- the centre of gravity of the load is as close to the loader as possible
- the load is kept as close to the ground as possible
- lifting with the fork tips is avoided
- Sudden changes in speed or direction are avoided

6.2. Driving with the forklift mast

When handling heavy loads, for the best traction and manoeuvrability, it must be made sure that enough weight is applied on the loader wheels. The amount of weight applied on the loader wheels can be influenced with the loader boom.

When lifting the loader boom slightly, more weight will be applied on the front wheels for better traction. The wheels of the forklift mast should not be lifted off from the ground, but with the boom a portion of the load can be transferred to the loader wheels. Lifting the boom too much will result in lifting the rear wheels of the loader off from the ground, leading to decreased or loss of manoeuvrability, as the loader tends to tip forward around the wheels of the forklift mast.

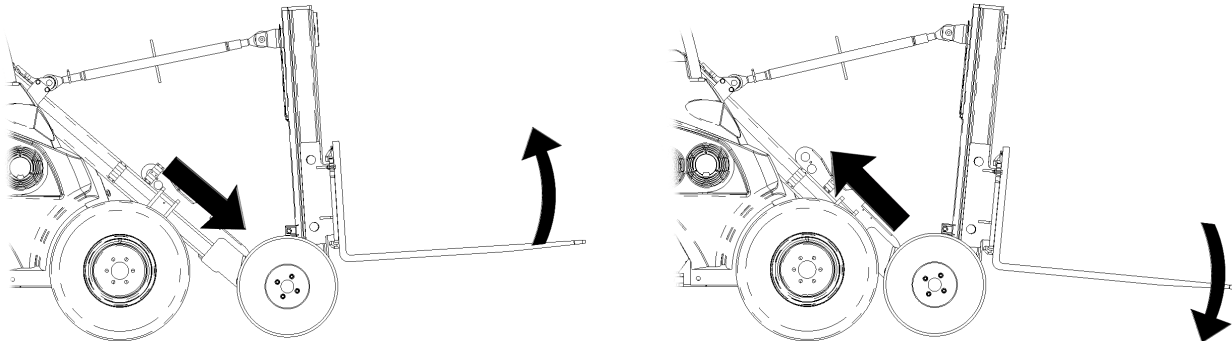
When the wheels of the forklift mast are on the ground, the turning radius of the loader will increase significantly, as the turning will be determined by the distance between the loader rear wheels and the wheels of the forklift mast. For this reason, the supporting bracket should be adjusted so that the forks are in horizontal position when the forklift mast is as close to the loader as possible. In transportation of an unloaded attachment its wheel can be lifted slightly off from the ground.



Practise the influence of the increased turning radius on an open area when driving with the forklift mast.

6.2.1. Adjusting the tilting of the forks

The tilting of the forklift mast can be done with the telescopic loader boom. When the supporting rod of the forklift mast is adjusted correctly, the forks are tilted slightly forward when the telescopic boom is fully retracted. When operating the telescopic boom, it may be necessary to operate the boom tilting and lifting as well to release the pressure and therefore allow the boom to move and tilt.



6.2.2. Driving on inclined terrains

Using the equipment on inclined terrains and slopes requires extra caution. Always keep the load as low and as close to the loader as possible. Drive slowly and do not make sudden changes in speed or direction. Operate the mast with careful and smooth movements. On inclined terrains the load should not be lifted high. Turning the articulation of the loader during lifting will affect the stability of the loader and may lead to overturning the machine on inclined terrains.

When transporting goods in uneven terrains or slopes, ensure, that the load is secured on the pallet and the forks. Use load securing straps when needed.

7. Maintenance and service

Continuous maintenance of the forklift mast includes, in addition to regular cleaning, monitoring the condition of its main components. Components, that should be monitored regularly, include hydraulic hoses and components, chain drive components and the condition of the forks.

Due to the crushing hazard caused by the moving parts of the forklift mast, all maintenance work must be performed while the forks are lowered against the ground. Most maintenance tasks can be performed while the forklift mast is attached to the loader. Shut down the loader and release the residual pressure.



Ensure that the attachment is properly supported before any maintenance work. Note that its parts may move also when the loader has been shut down and in particular watch out for pinching of fingers or hands.

Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Do not use the equipment, if you have discovered a fault in the hydraulic system. Leaking hydraulic fluid may penetrate the skin and cause serious injuries. Hydraulic fluid is also harmful to the environment.

Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of the hose has worn off. If there are signs of leaking, to check the component, hold up a piece of cardboard in the area where a leak is suspected. Finding any of these faults means that the hydraulic hose or component in question must be replaced. Spare parts are available from your nearest AVANT retailer or service point.



Never tighten a hydraulic fitting under pressure, since the fitting may break and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.



Repairing the pallet forks by welding is prohibited. Do not weld any hooks or other objects to the forks or to the frame. The forks are made of forged, heat-treated steel, and welding will have an adverse effect on their strength. If a fork gets bent, the damage may be permanent and the fork may not be straightened with normal measures. Replace a damaged fork with a new one.

7.1. Cleaning and lubricating the equipment

You may use a pressure washer and a mild detergent to clean the forklift mast. Do not use strong solvents or spray directly at the hydraulic components or at the labels on the attachment either. Before long storage periods touch up the paint finish or protect the visible metal surfaces by applying a small amount of oil on them.

Lubricate the chains and the two lubrication points near the quick attachment plate 2-3 times a year. Clean the nipple before applying grease. All lubrication nipples are standard R1/8" nipples, replace any damaged nipples. The bearings of the forklift mast are maintenance-free.

The forklift mast is equipped with replaceable slide pads, and for smoother operation they can be also lubricated if needed. The fork side shift should be used every now and then to prevent the jamming of the forks or the locking mechanisms. Check that the side locking of the forks is intact and locks in the intended slots. A suitable oil or grease may be used to make the fork side shift easier. Keep the frame securing bolts fastened. Contact AVANT service for spare parts.

8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) for attachment from the date of purchase.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work, such as the replacement of blades or the belts, or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus
EG-försäkran om överensstämmelse
EC Declaration of Conformity
EG-Konformitätserklärung



Valmistaja / Tillverkare / Manufacturer / Hersteller:

AVANT TECNO OY

Osoite / Adress / Address / Adresse:

Ylötie 1

33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveystaakumat (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:
 SFS-EN ISO 12100, SFS-EN ISO 4413

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats:
 SFS-EN ISO 12100, SFS-EN ISO 4413

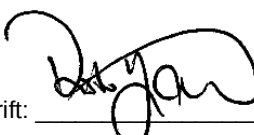
We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:
 SFS-EN ISO 12100, SFS-EN ISO 4413

Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen). Die folgende harmonisierte Normen wurden angewandt:
 SFS-EN ISO 12100, SFS-EN ISO 4413

Kuvaus: Avant-kuormainten kanssa käytettävä trukkimasto
 Beskrivning: Truckstativ för användning med Avant-lastare.
 Description: Forklift mast for use with Avant loaders.
 Beschreibung: Stapelmast für die Verwendung mit Avant Radlarer.

Mallit / Modeller / Models / Modellen:

Avant		400-700 series
Trukkimasto Truckstativ Forklift mast Stapelmst	175 cm, 1500 kg	A32915

Allekirjoitus / Namnteckning / Signature / Unterschrift:  1.2.2011 Ylöjärvi, Finland

Risto Käkälä,
 Toimitusjohtaja / Verkställande direktör /
 Managing Director / Geschäftsführer





AVANT[®]